



كلية تكنولوجيا الصناعة والطاقة
FACULTY OF INDUSTRIAL AND ENERGY TECHNOLOGY



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SOLAR TECHNOLOGY (Sol Tech)

A Multi-Functional Intelligent Unit for

Thermal Processing of Food Products

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By

Ahmed Rafek	Asmaa Nasr	Gaber Mohamed	Rawan AbdelHalim
Ahmed ElSaied	Alaa Salah	Ganna Mostafa	Sherouk Elsayed
Abdullah Ebrahim	Bassant Mostafa	Mai Saleh	Shahd Salah
Abdelrahman Khalid	Eman Ayman	Mariam Ayman	Shahd Mahmoud
Abdalrhman Eldeeb	Esraa Ahmed	Marim Khalid	Sama Osama
Akram Bahnasi	Fatma Elzahraa Mohamed	Mennatallah Magdy	Sama Omar Tasneem Saad

Under Supervision of

**Assoc. Prof. Amira M. Galal
Darwish**

Head of Food Industry Technology Program, *Faculty of Industrial and Energy Technology, Borg Al Arab Technological University (BATU).*

Dr. Tamer Rabie Elzeny
Lecturer at Food Industry Technology Program, *Faculty of Industrial and Energy Technology, Borg Al Arab Technological University (BATU).*

Dr. Arig Wahid Elkhoully
Lecturer at Food Industry Technology Program, *Faculty of Industrial and Energy Technology, Borg Al Arab Technological University (BATU)*

Eng. Shahd Hazem Tawfik
Demonstrator at Food Industry Technology Program, *Faculty of Industrial and Energy Technology, Borg Al Arab Technological University (BATU).*



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Submitted by

Serial	Student Name	Student ID	Serial	Student Name	Student ID
1	Ahmed Rafek	2450426	14	Mai Saleh	2450355
2	Ahmed ElSaied	2450005	15	Mariam Ayman	2450307
3	Abdullah Ebrahim	2450222	16	Marim Khalid	2450309
4	Abdelrahman Khalid	2450210	17	Mennatallah Magdy	2450336
5	Abdalrhman Eldeeb	2450218	18	Rawan AbdelHalim	2450151
6	Akram Bahnasi	2450043	19	Sherouk Elsayed	2450184
7	Asmaa Nasr	2450039	20	Shahd Salah	2450193
8	Alaa Salah	2450046	21	Shahd Mahmoud	2450197
9	Bassant Mostafa	2450082	22	Sama Osama	2450174
10	Eman Ayman	2450064	23	Sama Omar	2450175
11	Esraa Ahmed	2450029	24	Fatma Elzahraa Mohamed	2450430
12	Gaber Mohamed	2450084			
13	Ganna Mostafa	2450087	25	Tasneem Saad	2450083

Under Supervision of
Assoc. prof. Amira M. Galal Darwish
Dr. Arig Wahid Elkhoully
Dr. Tamer Rabie Elzeny
Eng. Shahd Hazem Tawfik



ABSTRACT

In order to meet the increasing need for effective, secure, and sustainable food processing systems, this project offers the design and development of a novel automated smart liquid pasteurization system that combines cutting-edge engineering concepts with contemporary control technology.

The system's foundation is jacketed heating technology, which is bolstered by an internal agitator to guarantee even heat transfer and avoid hot spot formation, protecting the processed liquid's sensory and nutritional qualities. In order to ensure efficient microbiological reduction and reliable product safety, the product goes through a regulated holding section after the heating stage, where it is kept at the necessary pasteurization temperature for a predetermined residence period. The product then goes through a cooling stage to lower its temperature, which helps to maintain the end product's quality, safety and shelf life while reducing the possibility of microbial development.

A fully integrated Cleaning-in-Place (CIP) system minimizes manual intervention and lowers contamination hazards in order to improve hygiene and operating efficiency. Additionally, the unit is powered by a hybrid energy system that combines conventional electrical supply with solar power, guaranteeing consistent and dependable functioning while lowering total energy consumption and encouraging environmental sustainability.

Real-time monitoring, data analysis, and intelligent process control are made possible by the system's fully automated operation, which integrates artificial intelligence (AI) and cutting-edge sensor technology. High accuracy, less human interaction, operational stability, and constant product quality are therefore guaranteed.



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LIST OF ABBREVIATIONS

HTST	High-temperature short-time
LTLT	Low-Temperature Long- Time
UHT	Ultra-High Temperature
CIP	cleaning-in-place
PLC	Programmable Logic Controller
STEC	Shiga toxin-producing Escherichia coli
HUS	hemolytic uremic syndrome
HMI	Human Machine Interface

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Chapter 1

LITERATURE REVIEW

A review of the literature reveals that research on dairy pasteurization and solar-assisted thermal systems has largely evolved in a compartmentalized manner, with individual studies focusing separately on hydrodynamics, heat-transfer enhancement, or energy integration, while lacking a unified, year-round validated pasteurization framework that simultaneously addresses seasonal solar variability, hybrid operational reliability, continuous-flow thermal behavior, and product safety under real climatic conditions; consequently, there is a clear need to move beyond short-term or component-level investigations toward a system-level approach capable of delivering regulatory-compliant pasteurization, particularly for small- and medium-scale dairy units in energy-constrained regions. To address this gap, the present study develops a hybrid solar-assisted automatic milk pasteurization system designed to effectively utilize high solar availability during peak radiation periods while ensuring continuous and reliable operation through auxiliary conventional energy support under low-solar conditions; the originality of this work lies in its comprehensive and integrated evaluation framework, combining long-term experimental monitoring, validated numerical simulations, detailed energy and exergy analyses, environmental impact assessment, techno-economic evaluation and microbiological validation of product safety within a single unified approach, thereby providing both scientific evidence and practical validation for advancing solar-assisted hybrid pasteurization systems toward scalable, reliable and economically viable deployment (Ray *et al.*, 2026).

1. Pasteurization

1.1. Pasteurization History

Thermal processing is one of the oldest and most widely used techniques for preserving food. It involves applying heat to food products to kill or inactivate harmful microorganisms that can cause spoilage or foodborne illnesses. This heat treatment also helps to extend the shelf life of food, making it safe for storage and consumption over extended periods (Tomasz, 2024).

A fundamental milestone in this field is pasteurization, which was developed in the nineteenth century by the French scientist Louis Pasteur, who initially applied the process to wine and beer in order to prevent spoilage. In 1862, the high-temperature short-time (HTST) approach was introduced for the first time, marking a significant advancement in food preservation. During the early twentieth century, rapid technological progress led to the development of various pasteurization methods. In 1908, Chicago implemented the



first law mandating the pasteurization of milk, highlighting the growing recognition of its importance for public health. Throughout the first half of the twentieth century, the holding method, also known as vat pasteurization (LTLT – Low Temperature Long Time), was the predominant technique; this method involved batch heating milk at approximately 63 °C for 30 minutes. By the early 1940s, however, continuous HTST systems had gained widespread acceptance in industrial applications. As pasteurization technology continued to advance, several combinations of temperature and holding time were established. Today, HTST pasteurization is widely recognized as the standard method for milk processing and is commonly performed using plate heat exchangers, while modern pasteurization plants incorporate multiple heating and cooling stages to ensure that milk achieves the required temperatures safely and efficiently (**Magar, 2023**).

1.2. Pasteurization Importance

Pasteurization systems are developed to achieve a 99.999% reduction of the microbial load by using the most heat-tolerant target pathogen *Coxiella burnetii*. The higher temperature is applied, the shorter the heat treatment time is required. The pasteurization time–temperature combinations were designed to eliminate harmful bacteria such as *Listeria*, *Salmonella*, *Campylobacter*, and *Mycobacterium bovis*, which posed serious health risks before the widespread application of pasteurization in commercial milk production (**Wittwer et al., 2022**).

1.3. Pasteurization Types

- Vat Pasteurization: Heating food to 145°F (63°C) for 30 minutes. Commonly used for milk. It's also known as Low-Temperature Long Time (LTLT) pasteurization.
- High-Temperature Short Time (HTST) Pasteurization: Heating to 161°F (72°C) for 15 seconds. This is the most common method for milk and fruit juice.
- Higher-Heat Shorter Time (HHST) Pasteurization: Similar to HTST but at higher temperatures, typically 192-212°F (89-100°C) for a few seconds.
- Ultra-High Temperature (UHT) Pasteurization: Heating to 275°F (135°C) for 2-5 seconds. Used for products needing a longer refrigerated shelf life, like cream and milk (**Goff and Griffiths, 2020**).

1.4. Solar Energy

The utilization of solar energy dates back to ancient civilizations, where it was applied in simple forms such as heating and drying. However, systematic scientific research and technological development of solar energy remained limited until the mid-twentieth century. Significant progress emerged following the energy crises of the



1970s, which intensified global interest in alternative and renewable energy sources. Since then, advancements in solar thermal collectors and energy conversion technologies have enabled the practical integration of solar energy into industrial processes, including heat-intensive operations such as milk pasteurization (Kumar and Prakash, 2024).

Despite their high efficiency in ensuring milk safety and quality, pasteurization systems are highly energy-intensive. They require substantial thermal and electrical energy for operations such as pasteurization, cleaning-in-place (CIP), mixing, pump operation, and control panel systems. To meet these demands, dairy plants rely heavily on conventional energy sources, including steam boilers, electric heaters, gas burners, or firewood-based technologies. The extensive dependence on fossil-fuel-based thermal energy not only increases operational costs but also contributes significantly to greenhouse gas emissions, indoor air pollution, and the depletion of non-renewable energy resources. Consequently, energy consumption in the dairy industry has become a major economic and environmental challenge, highlighting the urgent need for more sustainable heat generation solutions. Integrating renewable energy sources, particularly solar thermal technologies, offers a viable pathway to reduce energy costs, enhance process reliability, and minimize environmental impact. Solar energy is abundant, freely available, and widely accessible in many dairy-producing regions, making it an attractive alternative for supplying thermal energy to pasteurization processes. Moreover, when combined with hybrid or auxiliary energy systems, solar technologies can effectively mitigate challenges related to seasonal and diurnal intermittency. Therefore, the adoption of solar-based thermal energy systems represents one of the most practical and sustainable solutions for improving the energy efficiency and environmental performance of milk pasteurization processes (Ray *et al.*, 2026).

1.5.Pasteurization Device

1.5.1.Theory of operation

The pasteurization process operates through three fundamental stages designed to ensure food safety while maintaining product quality.

1. **Heating:** The product is heated to a predetermined temperature, which depends on the selected pasteurization method (typically ranging from 63°C to 135°C). This step aims to reduce or eliminate pathogenic microorganisms.
2. **Holding:** The product is maintained at the target temperature for a specific duration to ensure effective microbial inactivation.



3. **Cooling:** Finally, the product is rapidly cooled to inhibit the growth of any surviving microorganisms and preserve quality. This operating principle allows pasteurization systems to achieve microbial safety with minimal impact on nutritional and sensory properties, although certain high-temperature methods may affect flavor and nutrient content (Bylund, 2015; Toledo *et al.*, 2018).

Programmable Logic Controller (PLC)–based control systems enable real-time monitoring and precise regulation of critical process parameters, including temperature, processing time, and operating speed, thereby enhancing pasteurization performance, improving product quality, and significantly reducing the potential for human error (Kapireddy, 2025)

Aim of work

The core objective of this research is the development and validation of an innovative, energy-efficient pasteurization system that achieves strict microbial safety standards. Through the implementation of a hybrid solar thermal network coupled with real-time monitoring technology, this system mitigates environmental impact by displacing traditional fossil-fuel consumption in dairy processing.



Chapter 2

CASE STUDY/ PROBLEM STATEMENT

The SolTech Project has been developed to address several critical challenges through the implementation of integrated technological solutions, as outlined below.

2.1. PROBLEM

2.1.1. Temperature Fluctuation.

Maintaining precise temperature control is a core responsibility for food processing operations (**Keithley, 2025**), as the extent of microorganism inactivation depends on the combination of temperature and holding time. Accordingly, minimum temperature and time requirements for milk pasteurization are established based on thermal death time studies of the most heat-resistant pathogen found in milk (**Goff et al., 2020**). However, temperatures exceeding the standard pasteurization limits may contribute to increased nutrient degradation, highlighting the importance of maintaining temperature within a precise range to ensure both safety and quality (**Pasteurisation and Food Safety, 2022**).

In practice, however, maintaining this precise control can be challenging, as temperature control failures in dairy processing often occur at critical control points due to reliance on manual monitoring methods, which are unable to capture real-time variations. In addition, human errors in recording and interpreting data further increase the likelihood of undetected deviations, leading to microbial growth, product spoilage, and non-compliance. Automated sensor-based monitoring systems, however, provide continuous real-time data, enabling immediate detection of deviations and ensuring consistent control of critical parameters, thereby improving product safety and operational efficiency (**Nb, 2026**).

2.1.2. Effect of Pasteurization Temperature and Holding Time Below Standard Levels.

Inadequate pasteurization conditions may allow pathogenic bacteria present in raw milk to survive, thereby compromising product safety (**Oehler, 2025**). Consuming milk that has been inadequately pasteurized due to process failures can expose individuals to harmful pathogens such as *Campylobacter*, *Cryptosporidium*, *Escherichia coli*, *Listeria*, *Brucella*, and *Salmonella*. Certain groups, including children under five, adults over 65, pregnant women, and immunocompromised individuals, are at a higher risk of developing severe illness from these pathogens. In particular, *Listeria monocytogenes* poses a significant risk to pregnant women, as it can lead to miscarriage, severe illness, or even death of the newborn, and may harm the fetus even when the mother shows no symptoms.



Symptoms of foodborne illness associated with inadequately pasteurized milk may include diarrhea, stomach cramps and vomiting; however, more serious complications can also occur (Marotta *et al.*, 2026).

For instance, *Campylobacter* infection has been linked to Guillain–Barré syndrome, a condition in which the immune system attacks the nerves, leading to muscle weakness and, in some cases, paralysis. Additionally, infections caused by Shiga toxin-producing *Escherichia coli* (STEC) may result in hemolytic uremic syndrome (HUS), a severe condition that can cause kidney failure, long-term health complications, or even death (Wysok *et al.*, 2024).

2.1.3. Effect of Pasteurization Temperature and Holding Time Above Standard Levels.

Milk has high nutritional value and contains nutrients such as proteins, fats, lactose, essential fatty acids, vitamins, and enzymes that contribute to normal human growth, development, and body functions). Thermal treatment of milk, including pasteurization and ultra-high temperature (UHT) processing, is essential for ensuring microbiological safety and extending shelf life; however, the intensity of heating has a significant impact on milk quality. Increased temperatures lead to structural changes in milk proteins, particularly the denaturation of whey proteins and their interaction with casein micelles, which may reduce protein functionality and bioavailability (Bai *et al.*, 2023).

In addition, higher heat exposure can decrease the retention of heat-sensitive vitamins and micronutrients, especially when processing conditions are not well controlled. Thermal processing also promotes the formation of heat-induced compounds such as Maillard reaction products, lactulose, and furosine, which are commonly used as indicators of processing severity and nutritional damage. Therefore, although moderate heat treatments preserve most of the nutritional value of milk, excessive heating can negatively affect protein structure, nutrient content, and overall quality, making proper control of time–temperature conditions essential to maintain both safety and nutritional integrity (Wang *et al.*, 2024).

2.2.Sol Tech Solution

As mentioned before pasteurization is a critical step in food processing, as it ensures both product safety and quality. In modern systems like our project, smart sensors play a vital role in optimizing this process by continuously monitoring and regulating the key parameters required for effective pasteurization. The integration of sensor technologies allows precise control of variables such as temperature and holding time, ensuring that each product is processed under optimal conditions without compromising safety or



quality. Furthermore, advanced sensor systems—including fast-response temperature sensors and automated control units—enable real-time adjustments, thereby minimizing process deviations and maintaining consistent product standards.

2.2.1.Solving Energy Shortage and Environmental Pollution Using Solar Energy

Burning fossil fuels increases the concentration of greenhouse gases such as carbon dioxide and methane in the atmosphere, which strengthens the greenhouse effect and leads to a faster rise in global temperatures. This results in serious environmental impacts like melting glaciers, rising sea levels, and more frequent extreme weather events such as floods, cyclones, heat waves, and droughts. The Sol-Tech device, which uses solar energy as a source in pasteurization, provides a sustainable alternative to these harmful methods by relying mainly on clean solar power that produces no greenhouse gas emissions. This hybrid system, where solar energy is used with conventional electricity, ensuring continuous operation. Unlike fossil fuels, which are limited and contribute heavily to pollution, solar energy is renewable and abundant, allowing it to meet energy needs while reducing environmental damage. A carbon footprint refers to the amount of greenhouse gases released due to human activities, and fossil fuels are the main cause of it. In contrast, solar energy is one of the cleanest energy sources, as it significantly reduces carbon dioxide emissions and helps mitigate climate change. Therefore, integrating solar technology in devices like Sol-Tech not only reduces pollution but also provides a reliable energy solution, making it an effective approach to addressing both energy shortages and environmental challenges (Bolt, 2023).

2.2.2.Sol Tech Smart automated pasteurization

The use of an automated system plays a key role in reducing operational costs by minimizing the need for manual labor. Automation allows processes to run efficiently with fewer workers, which lowers expenses and increases productivity. In addition, automated systems help reduce human errors, ensuring greater accuracy and consistency in operation. As a result, the system becomes more reliable, efficient and suitable for sustainable and large-scale applications.



Chapter 3 METHODOLOGY

This chapter presents the methodology adopted for the design and implementation of the smart pasteurization unit. The system is structured into several subsystems, including the vessel, heating system, Agitator, solar energy system, and control unit.

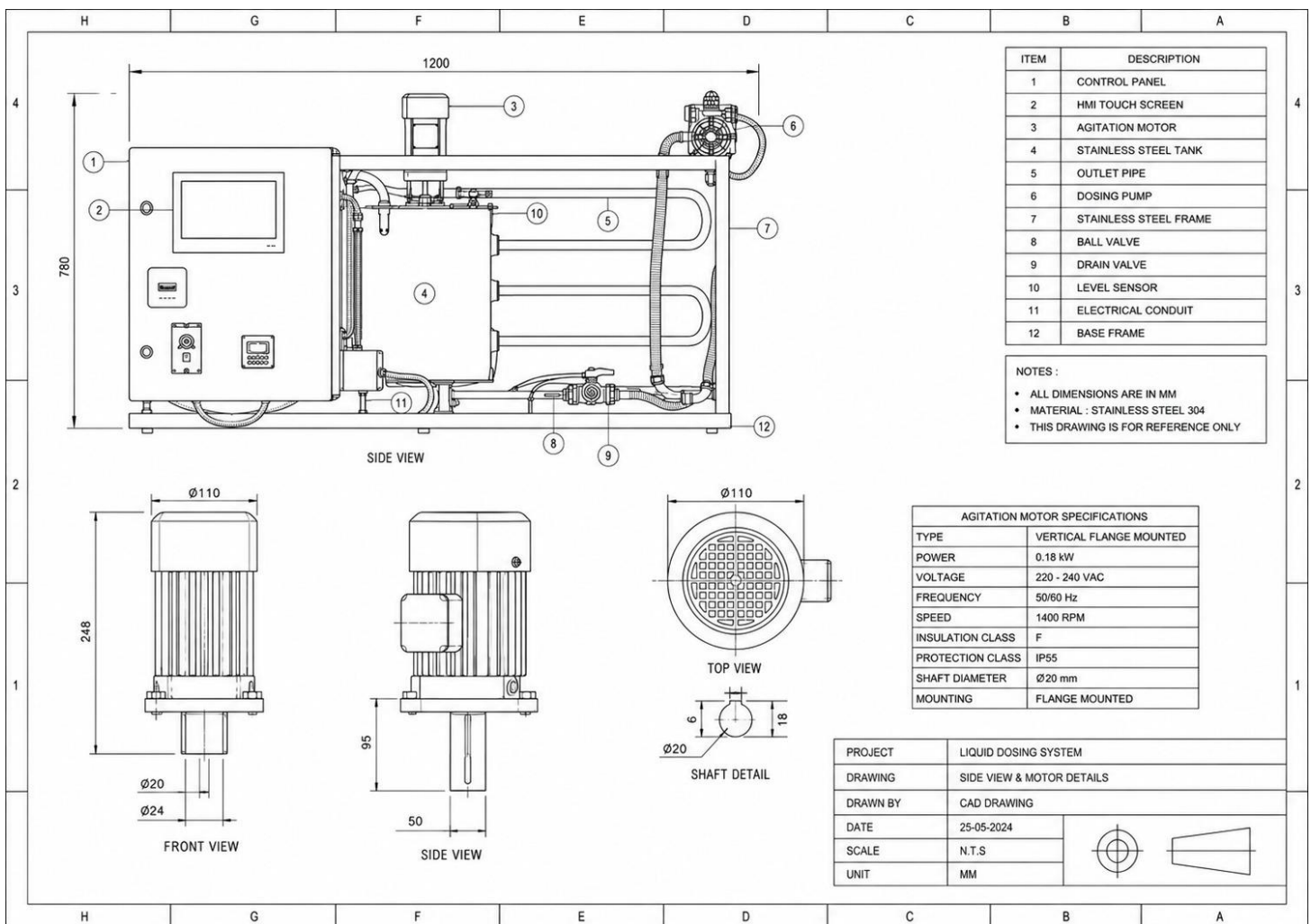


Fig.1. AutoCAD-generated engineering drawing of the pasteurization unit, illustrating the overall system design, arrangement of major components, and key dimensional specifications.



Fig. 2. Three-dimensional (3D) model of the designed pasteurization unit.

3.1.Stainless Steel double jacket Vessel Fabrication Process:

Stainless steel is one of the most important raw materials in stainless steel tank manufacturing. In addition to its excellent corrosion resistance, it offers a hygienic surface, high mechanical strength, and precise machinability, while its aesthetics and chemical neutrality also make it a common choice. The processing of stainless steel in tank production can be divided into four main areas: cutting, forming, welding, and surface treatment, each of which involves different methods depending on the application. (Károly, 2025) For food processing applications such as pasteurization, material selection is critical to ensure hygiene, resistance to corrosion, and ease of cleaning. Among the various grades of stainless steel, Grade 304 is one of the most commonly used due to its excellent balance of properties. It is an austenitic alloy composed of 18% chromium and 8% nickel, which provides high resistance to oxidation, chemicals, and moisture. In addition, its smooth and non-reactive surface makes it ideal for food-contact applications, while its high ductility facilitates forming processes such as rolling and shaping during tank fabrication. Its non-magnetic nature in the annealed condition also

contributes to its suitability in industrial food processing systems (Al-Khafaji and Santos, 2025).

3.1.1.Cutting

The initial step in stainless steel fabrication is cutting the raw material into the desired shapes and sizes. In this work, laser cutting was exclusively used due to its high precision and ability to produce clean, accurate edges with minimal need for post-processing. This method utilizes a high-powered laser beam, making it particularly suitable for achieving intricate geometries and fine details. The fabricated component in this project was a double-jacket stainless steel vessel with a capacity of 20 liters, designed according to the required process specifications (**Comprehensive Guide to Stainless Steel Fabrication: A Step-By-Step Process - MFG Shop, 2025**)

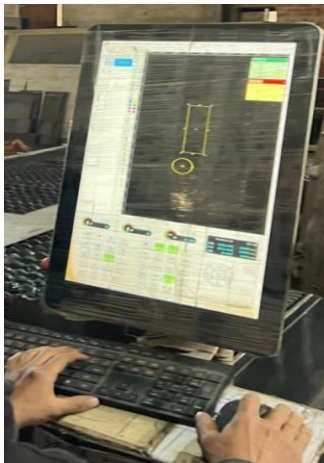


Fig. 3: CAD design and laser cutting program setup for the stainless steel vessel components.



Fig. 4: Laser cutting process of stainless steel sheets used for fabricating the vessel components.

3.1.2. Bending and Forming (Rolling)

After the cutting stage, the stainless steel sheets were subjected to forming processes. In our project, rolling was used to convert the flat sheets into cylindrical forms required for the vessel body and the outer jacket. This process was carried out using rolling machines, allowing the material to achieve the desired curvature without pre-heating while maintaining dimensional accuracy and material properties (**Comprehensive Guide to Stainless Steel Fabrication: A Step-By-Step Process - MFG Shop, 2025**). A

double-jacket configuration was then achieved by assembling two concentric shells, creating a space between them for heat transfer fluid circulation (Connor, 2024).



Fig. 5: Rolled stainless steel sheet during fabrication of the vessel body.

3.1.3. Welding and Surface Treatment

TIG welding was used to join the stainless steel components of the inner vessel and the outer jacket, ensuring strong, clean, and leak-proof joints required for the double-jacket configuration. Following welding, surface treatment was carried out to improve surface quality and corrosion resistance. This involved cleaning and polishing operations to remove weld residues and achieve a smooth, hygienic finish, which is essential for safe and efficient operation in food processing applications such as pasteurization (Comprehensive Guide to Stainless Steel Fabrication: A Step-By-Step Process - MFG Shop, 2025).



Fig. 6. Mechanical polishing of the stainless steel surface after welding.



3.1.4.Heater

An electrical immersion heater operating at 220 V is used as the main heat source in the system. Immersion heaters are designed to heat liquids by being inserted directly into the heating medium. The heater consists of tubular heating elements mounted on a flange, allowing it to be installed inside the jacket surrounding the vessel. It operates based on electrical resistance heating (Joule effect), where heat is generated when electric current passes through a resistive element. The heat produced is transferred to the heating medium inside the jacket, which then transfers heat to the product in the inner vessel through indirect heating. This heating approach enables the system to achieve and maintain the required pasteurization temperature for a specified period, ensuring effective microbial reduction while preserving the nutritional and sensory qualities of the product. Immersion heaters are widely used in liquid heating applications due to their high efficiency, uniform heat distribution, and reliable performance. In this system, using a 220 V heater ensures compatibility with standard electrical supply and ease of operation (Immersion Heater: The Complete Guide, n.d).

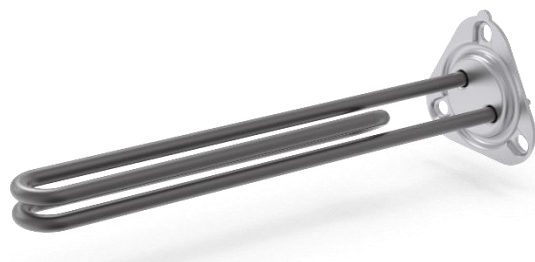


Fig.7: Tubular immersion heater installed in the jacketed vessel.

3.1.5. Agitator

The agitator is driven by a 5IK120RGU-CF single-phase speed-control induction motor rated at 120 W and 220 V. The motor operates at a speed of 1300–1500 rpm and provides the mechanical power required to drive the stainless-steel paddle agitator. During the pasteurization process, the motor ensures continuous mixing of the product, which enhances heat transfer and promotes uniform temperature distribution throughout the vessel. This prevents the formation of hot spots and ensures that all portions of the liquid are exposed to the required pasteurization temperature, thereby improving microbial reduction and maintaining product quality [Hot Item] 5ik120rgu-CF/5gu (X) K+Us52 120W 220V 50Hz/60Hz Single Phase AC Speed Control Motor With Controller, n.d.]



Fig.8. 5IK120RGU-CF speed-control induction motor used for driving the agitator



Fig.9. Stainless-steel paddle agitator used to provide uniform mixing during the pasteurization process

3.1.6.Cleaning-in-place (CIP)

The Cleaning-In-Place (CIP) system is integrated into the pasteurization unit to ensure effective cleaning and sanitation of all product-contact surfaces without requiring equipment disassembly. The system incorporates a stainless-steel static spray ball nozzle that provides 360° cleaning coverage through strategically distributed perforations, ensuring effective distribution of cleaning and sanitizing solutions over all internal tank surfaces. The spray ball is characterized by its corrosion-resistant construction and maintenance-free operation due to the absence of moving parts, making it suitable for hygienic food-processing applications (**European Hygienic Engineering & Design Group, 2022**).

The CIP process is considered a critical operation in dairy and pasteurization systems, as improper cleaning may lead to product spoilage, microbial contamination, equipment damage, and reduced process efficiency. A typical CIP cycle begins with a pre-rinse using cool water to remove residual product, dissolve sugars, and loosen deposits from the equipment surfaces. This is followed by an alkaline cleaning stage, commonly using sodium hydroxide (NaOH) at a concentration of 0.5–2.0%, to remove fats, proteins, and other organic residues accumulated during processing. An intermediate water rinse is then performed to remove any remaining alkaline solution before the acid cleaning stage. The acid rinse, typically using nitric acid or phosphoric acid at a concentration of 0.5–1.0%, is carried out to eliminate mineral deposits, scale, and milk stone that cannot be removed by alkaline detergents alone. Afterward, a final hot-water

rinse or sanitization step is carried out to remove residual cleaning chemicals and reduce microbial contamination before the next production cycle. The effectiveness of the CIP process depends on proper control of temperature, chemical concentration, contact time, and flow conditions, ensuring high hygiene standards, food safety, and reliable operation of the pasteurization system (**Kumar, 2026**).



Fig.10.Stainless-steel static spray ball nozzle used in the CIP system for tank cleaning and sanitization.

3.1.7.Pump and cooling break

The cooling section consists of a stainless-steel serpentine cooling coil integrated into the pasteurization system. The coil is designed to provide an extended heat transfer surface area, allowing efficient removal of heat from the pasteurized product after the holding stage. During operation, the product is transferred through the cooling coil by a centrifugal pump at a controlled flow rate while being exposed to a cooling medium, resulting in a gradual reduction in temperature. The pump ensures continuous product movement through the cooling section, thereby enhancing the efficiency of the cooling process. Rapid cooling after pasteurization is essential for maintaining product quality, preserving nutritional and sensory properties, and minimizing microbial growth. In addition, the stainless-steel construction ensures hygienic operation and compatibility with food processing applications.



Fig.11.Serpentine cooling coil designed to increase heat transfer area and improve cooling efficiency



Fig.12. Centrifugal pump used for product transfer within the pasteurization system

3.1.8. Programmable Logic Controller (PLC Unit) and Human Machine Interface (HMI).

A Programmable Logic Controller (PLC) serves as the core processing hub in industrial automation, continuously scanning input signals from field sensors and executing deterministic logic algorithms with microsecond-level response times to precisely regulate actuators, motors, and valves. To facilitate operator interaction with this automated environment, a Human-Machine Interface (HMI) acts as a graphical platform that visualizes real-time process data, handles alarm management, and transmits user-defined setpoints back to the controller. In modern industrial setups, especially where space constraints and architectural complexity are critical design factors, the utilization of integrated PLC+HMI combination devices has emerged as a highly efficient solution. By consolidating both real-time control processing and graphical user interface functionalities into a single compact unit, these integrated systems significantly minimize installation complexity, lower manufacturing costs, and streamline communication protocols at the machine level (S. Team, 2025).



Fig.13. Programmable Logic Controller (PLC)



Fig.14. Human Machine Interface (HMI)

3.1.9. Temperature Control System

During the pasteurization process, the ENDA ET4420 PID Temperature Controller (ET4420) was used to monitor and control the processing temperature. The controller receives temperature measurements from temperature sensors, including Pt100 RTD and thermocouple sensors (J, K, L, T, S, and R types), and compares the measured temperature with the preset pasteurization temperature. Based on the difference between the two values, the controller automatically adjusts the heating process to maintain the desired temperature. The PID control function provides accurate and stable temperature regulation, ensuring proper pasteurization conditions and consistent product quality. [ET4420 - Temperature Controllers - Automation - ENDA, n.d.]



Fig.15. ENDA ET4420 PID Temperature Controller

3.1.10.Solar Energy

The pasteurization system incorporates a photovoltaic (PV) solar energy subsystem as a supplementary power source. A solar panel with dimensions of 50×60 cm is used to convert solar radiation into electrical energy through photovoltaic cells. When exposed to sunlight, the PV cells generate electrical charges that produce usable electricity to support the operation of the system. The integration of solar energy reduces dependence on conventional electrical power, decreases overall energy consumption, and enhances the environmental sustainability of the pasteurization unit. In addition, the hybrid configuration combining solar power with conventional electrical supply ensures continuous and reliable operation under different operating conditions, **(according to U.S. Department of Energy. (2026)).**



Fig.16.Solar panel (50×60 cm).

3.2.Microbial Plate Count

The microbial count of the samples was determined using the pour plate method as a quality control measure to verify the effectiveness of the pasteurization process and ensure the microbiological quality of the final product. Liquid samples were serially diluted using sterile broth to obtain a countable microbial population within the acceptable range of 30–300 CFU/mL. For each selected dilution, 1 mL of the sample was transferred into a sterile Petri



dish, followed by the addition of approximately 15 mL of molten agar medium cooled to 40–45°C. The plate was gently swirled to ensure uniform distribution of the sample throughout the medium and then allowed to solidify. After solidification, the plates were incubated in an inverted position at 37°C for 24 hours. Following incubation, the developed colonies were counted, and the total microbial population was expressed as colony-forming units per milliliter (CFU/mL) (Dahal, 2025).

3.3.Product Flow

The process begins with loading the product into the processing tank. The product is then heated indirectly through the jacketed vessel until the required pasteurization temperature is achieved. Once the target temperature is reached, the product is maintained in the holding section for the specified residence time to ensure effectiveness of the pasteurization process. After the holding stage, the product passes through the cooling section, where its temperature is reduced to the desired level before discharge. Upon completion of the production cycle, the system undergoes a Cleaning-In-Place (CIP) procedure, during which cleaning solutions are circulated through the vessel and pipelines to prepare the unit for the next operation.

3.3.1.Specific Product (Iced Coffee)

Iced Coffee establishing a standardized baseline for subsequent thermal processing and pasteurization evaluation (Gomez & Harrison, 2026).

Table 1. Low-Fat Iced Coffee Beverage Ingredients.

Ingredient	Quantity	Function
Low-Fat Milk	900 mL	Base ingredient
Cold Brew Coffee Extract	80 mL	Coffee flavor
Date Syrup (or Stevia)	20 mL	Natural sweetener
Vanilla Extract (Optional)	2 mL	Flavor enhancement



Table.2. Processing Parameters of the Low-Fat Iced Coffee Beverage.

Process Step	Condition
Mixing	3–5 min
Pasteurization Temperature	72°C
Holding Time	15 sec
Rapid Cooling	$\leq 5^{\circ}\text{C}$
Filling Temperature	4–5°C
Storage	Refrigerated (2–5°C)

3.3.2. Quality Parameters

Table.3. Quality Parameters of the Low-Fat Iced Coffee Beverage

Quality Parameters	Specification
Appearance	Light brown, homogeneous
Flavor	Mild coffee with creamy taste
Odor	Fresh coffee aroma
pH	6.5 – 6.8
Shelf Life	5–7 days (Refrigerated)



Chapter 4

EXPERIMENTAL RESULTS

4.1. Estimated results of Pasteurization according to the following

Table.4. Experimental Results of the Pasteurization Process

Product	Processing Type	Temperature	Time	Final Product	References
Milk	HTST Pasteurization	72°C	15 sec	Pasteurized Milk	<i>FDA PMO (Pasteurized Milk Ordinance).</i>
Milk	LTLT Pasteurization	63°C	30 min	Pasteurized Milk	<i>Codex Alimentarius / Dairy Processing Handbook.</i>
Fruit Juice	Pasteurization	85–95°C	15–30 sec	Pasteurized Juice	<i>Fellows, P. J. (Food Processing Technology).</i>
Cream	Pasteurization	75–85°C	15–20 sec	Processed Cream	<i>Tetra Pak Dairy Processing Handbook</i>
Ice Cream Mix	Pasteurization	80–85°C	15–30 sec	Pasteurized Ice Cream Mix	<i>Goff & Hartel (Ice Cream textbook).</i>
Vinegar	Mild Pasteurization	60–70°C	10–15 min	Stabilized Vinegar	<i>Jay, J. M. (Modern Food Microbiology)</i>
Tomato Sauce	Pasteurization	85–95°C	5–10 min	Ready-to-Use Sauce	<i>Preservation of Foodstuffs (Thermal Processing</i>



Chapter 5

COST ESTIMATION

5.1. The estimated cost of the designed system is as follows:

Table.5. Estimated Manufacturing Cost of the Pasteurization Unit

Parts	Components	Price (EGP)
Structure	- Stainless steel body - Heat exchanger pipes - Manufacturing & Fabrication	23,260
Electrical System	- PLC - Power supply - Control panel (PVC) - Heater - Mixer motor - Solenoid valve - Temperature controller - HMI - Cable connections - Insulation & connections - Signal lamp - Selector switch - Power switch - Relay contacts - Speed controller - Sound system	35,195
Plumbing	- Hose clamps - Hoses - Pump - Nipple - Ferrule - Connectors	3,380
Solar Energy	Solar panels	4,000
Total		66,335



Chapter 6

CONCLUSION & FUTURE WORK

Despite the successful performance of the developed system, several improvements can be implemented in future research to enhance its functionality, efficiency, and sustainability. The following recommendations are proposed for further development of the pasteurization unit:

1. Complete Cooling System

Replace the current cooling coil with a complete refrigeration system utilizing a plate heat exchanger or chiller to achieve faster and more efficient cooling, thereby improving product quality and reducing processing time.

2. Full Solar Energy Integration

Operate the entire unit using solar energy through photovoltaic panels, batteries, and an inverter system to reduce dependence on conventional electricity and enhance environmental sustainability.

3. Artificial Intelligence Integration

Integrate Artificial Intelligence (AI) algorithms to optimize heating and cooling operations, predict processing parameters, monitor product quality, detect system faults, and enable predictive maintenance.

4. Remote Monitoring and IoT Integration

Connect the unit to the Internet of Things (IoT) for real-time monitoring and remote control through a mobile application or web-based dashboard. This would allow users to track temperature, process parameters, alarm conditions, and overall system performance from any location.

5. Energy Recovery System

Implement a heat recovery system to capture and reuse thermal energy generated during processing for preheating water or other utility streams. This would reduce overall energy consumption and improve system efficiency.



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In conclusion, this project successfully demonstrated the design and implementation of a Smart Multi-Functional Thermal Processing Unit that integrates the key stages of food thermal processing into a single compact and efficient system. By combining indirect heating, continuous mixing, rapid cooling, automated Clean-in-Place (CIP) cleaning, PLC-based intelligent control, and solar energy integration, the developed unit enhances processing efficiency, ensures food safety, maintains product quality, improves operational reliability, and promotes energy sustainability. The obtained results confirm the feasibility and effectiveness of the proposed design in providing precise process control and reliable performance while reducing operational costs and energy consumption. Therefore, the system represents a practical, scalable, and cost-effective solution for small-scale food processing industries, research facilities, and educational applications, while also serving as a strong foundation for future developments in smart and sustainable food processing technologies.



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الملخص العربي

تستهدف هذه الدراسة تصميم، وتطوير، وتقييم كفاءة وحدة نمذجة ذكية ومتعددة الوظائف للمعالجة الحرارية وبسترة الأغذية السائلة، من خلال تكامل الهندسة الميكانيكية مع أنظمة التحكم الآلي المتقدمة لضمان تحقيق السلامة الميكروبيولوجية بكفاءة تشغيلية عالية وتقليل استهلاك الطاقة. تعتمد الآلية الديناميكية للوحدة على نمط التسخين غير المباشر عبر غلاف مائي مزدوج (**Double-Jacketed Vessel**) مدعوم بنظام تقليب ميكانيكي (**Mechanical Agitation System**)؛ وذلك لضمان التوزيع المتجانس للمستويات الحرارية داخل الكتلة الحيوية للمنتج، وتحييد ظاهرة الإجهاد الحراري الموضعي (**Thermal Hotspots**)، مما يحافظ على القيمة التغذوية والخصائص الأورجانبولبتية (الحسية) للمادة المعالجة. وتتبع العملية الحرارية بروتوكول البسترة القياسي المتمثل في ثلاث مراحل متتابعة تشمل: رفع درجة الحرارة الحرجة، والاحتفاظ الحراري المؤقت (**Holding Time**) للتحقق من حركية الموت الحراري للميكروبات المستهدفة، تليها مرحلة التبريد السريع لخفض الحمل الحراري وتثبيت التفاعلات الكيميائية غير المرغوبة. ولتأصيل معايير سلامة الغذاء ومطابقة الاشتراطات الصحية الصارمة، تم دمج نظام التنظيف الموضعي التلقائي (**Cleaning-In-Place / CIP**) كآلية تطهير مغلقة ومستدامة، تضمن تعقيم كافة الأسطح البيئية الملامسة للمنتج دون الحاجة لتفكيك النظام الميكانيكي، مما يسهم في خفض زمن التعطيل التشغيلي ورفع الكفاءة الإنتاجية الإجمالية.

تدار العمليات التشغيلية للوحدة عبر منظومة كهروميكانيكية متكاملة للتحكم الآلي، تركز على وحدة التحكم المنطقي المبرمج (**PLC**) الموصولة بشاشة واجهة المستخدم الرسومية (**HMI**)، بالإضافة إلى شبكة من المستشعرات الدقيقة (**Sensors**) والمحولات لمراقبة حركية درجات الحرارة والتحكم الذكي في المراحل المتتابعة للتشغيل تلقائياً. ولدعم استدامة النظام، تم دمج منظومة طاقة هجينة (**Hybrid Power System**) توفيق بين الخلايا الفوتوفلطية الشمسية ومصدر التيار الكهربائي التقليدي، مما يقلل البصمة الكربونية ويخفض استهلاك الطاقة مع ضمان استمرارية التشغيل تحت مختلف الظروف المناخية. وقد كشفت نتائج التشغيل الأولية ونمذجة الأداء عن استقرار هيدروليكي وحراري ممتاز للوحدة، محققة دقة عالية في ضبط درجات الحرارة مع توزيعها المتجانس وتأمين الخصائص الفيزيائية والكيميائية للمنتج النهائي، مما يثبت كفاءة وموثوقية التصميم المقترح. وتعد هذه الوحدة الذكية نموذجاً تطبيقياً قابلاً للتطوير مستقبلياً من خلال دمج دورات تبريد ميكانيكية متكاملة، والانتقال الكامل نحو الإمداد بالطاقة الشمسية المستقلة، فضلاً عن إدماج خوارزميات الذكاء الاصطناعي (**AI-Driven Control**) وأنظمة استرجاع الحرارة المفقودة (**Energy Recovery Systems**) تماشياً مع متطلبات الثورة الصناعية الرابعة في تكنولوجيا الأغذية المستدامة".



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APPENDICES



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Description:

" A Multi-Functional Thermal Processing Unit Using Solar Energy is an advanced system designed for processing liquid and semi-liquid products such as milk, juices, and sauces. The unit mainly performs pasteurization, heating, and cooling processes to maintain product quality and extend shelf life. It is constructed from stainless steel according to food safety standards and utilizes solar energy to reduce electricity consumption, lower operating costs, and improve energy efficiency and sustainability "

Team Members:

- | | | |
|-----------------------|-----------------------|----------------------------|
| 1. Ahmed Rafek | 10. Eman Ayman | 19. Rawan AbdelHalim |
| 2. Ahmed ElSaled | 11. Esraa Ahmed | 20. Sherouk Elsayed |
| 3. Abdullah Ebrahim | 12. Tasneem Saad | 21. Shahd Salah |
| 4. Abdelrahman Khalid | 13. Gaber Mohamed | 22. Shahd Mahmoud |
| 5. Abdalrhman Eldeeb | 14. Ganna Mostafa | 23. Sama Osama |
| 6. Akram Bahnasi | 15. Mai Saleh | 24. Sama Omar |
| 7. Asmaa Nasr | 16. Mariam Ayman | 25. Fatma Elzahraa Mohamed |
| 8. Alaa Salah | 17. Marim Khalid | |
| 9. Bassant Mostafa | 18. Mennatallah Magdy | |

Supervision
Dr. Arig Wahid
Dr. Tamer El-Zeny

Head of Project

Assoc. Prof. Amira M. Galal Darwish

Faculty Dean
Prof. Alaa Arafa

Supervision
Eng. Shahd Hazem

University President
Prof. Mohamed M. El-Gohary

Fig.17. SOLTECH Project Banner



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